

1 Solution — GIAM 3.1

Problem 3

1.1 Problem

Prove that the sum of two rational numbers is a rational number.

1.2 Statement

Written as an explicit conditional (the habit of Problem 1):

$$\forall x, y, \quad (x \in \mathbb{Q}) \wedge (y \in \mathbb{Q}) \implies (x + y \in \mathbb{Q}).$$

This is the **closure of $\mathbb{Q}$ under addition**.

1.3 Proof

***Proof.** Let x and y be arbitrary rational numbers. By the definition of $\mathbb{Q}$ there are integers a, b, c, d with $b \neq 0$ and $d \neq 0$ such that $x = a/b$ and $y = c/d$. By algebra,*

$$x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}.$$

Now $ad + bc$ is an integer, since $\mathbb{Z}$ is closed under multiplication and addition. Also bd is an integer, and $bd \neq 0$ because $b \neq 0$, $d \neq 0$, and $\mathbb{Z}$ has no zero divisors. So $x + y$ is a ratio of two integers with nonzero denominator, and therefore $x + y \in \mathbb{Q}$. Since x and y were arbitrary, the sum of any two rationals is rational. ■

1.4 The Shape of the Argument

Unpack both hypotheses into raw integers, compute, then repack the answer into the definition of $\mathbb{Q}$.

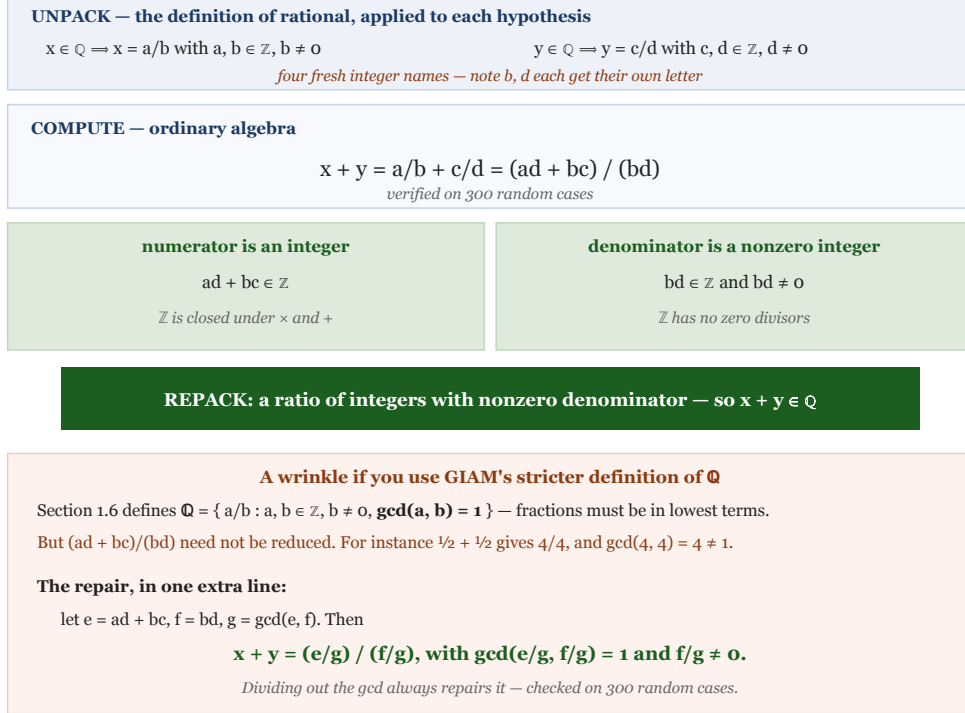


Figure 1.1: Top panel, UNPACK: the definition of rational applied to each hypothesis — x in $\mathbb{Q}$ gives x equal to a over b with a and b integers and b nonzero; y in $\mathbb{Q}$ gives y equal to c over d with c and d integers and d nonzero. Four fresh integer names are introduced, with b and d each getting their own letter. Second panel, COMPUTE: ordinary algebra gives x plus y equals a over b plus c over d equals $(ad$ plus $bc)$ over bd , verified on 300 random cases. Two green panels check the obligations: the numerator ad plus bc is an integer because the integers are closed under multiplication and addition; the denominator bd is an integer and is nonzero because the integers have no zero divisors. A REPACK banner concludes that x plus y is a ratio of integers with nonzero denominator, so x plus y is rational. A final panel describes a wrinkle if one uses GIAM's stricter Section 1.6 definition requiring gcd of numerator and denominator to be 1: the expression $(ad$ plus $bc)$ over bd need not be reduced — for instance one half plus one half gives four fourths, whose gcd is 4. The repair takes one extra line: set e equal to ad plus bc , f equal to bd , and g their gcd; then x plus y equals $(e$ over $g)$ over $(f$ over $g)$, with gcd of those equal to 1 and the denominator nonzero. Checked on 300 random cases.

The proof follows exactly the **unpack — compute — repack** skeleton identified in Problem 2.7.5:

1. **Unpack** — apply the definition of “rational” to each hypothesis, producing raw integers.

2. **Compute** — combine them with ordinary algebra.

3. **Repack** — verify the result satisfies the definition of “rational.”

The two checks in step 3 are the substance: the numerator must be an **integer**, and the denominator must be a **nonzero integer**.

1.5 Remark — A Wrinkle From the Stricter Definition of $\mathbb{Q}$

The proof above uses the informal definition $\mathbb{Q} = \{a/b : a, b \in \mathbb{Z}, b \neq 0\}$. But Section 1.6 replaced this with the *improved* version

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0, \gcd(a, b) = 1 \right\},$$

requiring fractions to be in **lowest terms** so that each rational has exactly one representative. Under that definition the proof is not quite finished: $(ad + bc)/(bd)$ need not be reduced. Concretely, $\frac{1}{2} + \frac{1}{2}$ gives $4/4$, and $\gcd(4, 4) = 4 \neq 1$.

The repair costs one line. Put $e = ad + bc$, $f = bd$, and $g = \gcd(e, f)$. Then

$$x + y = \frac{e/g}{f/g}, \quad \gcd\left(\frac{e}{g}, \frac{f}{g}\right) = 1, \quad \frac{f}{g} \neq 0,$$

so the sum has a lowest-terms representative and lies in $\mathbb{Q}$ on the strict definition too. (Verified on 300 random cases.) Noticing that the stated definition changes what still needs proving is exactly the kind of care Chapter 3 is training.

1.6 Remark — Where Each Hypothesis Is Used

Neither hypothesis is idle, and it is worth tracking them:

- $b \neq 0$ and $d \neq 0$ are needed twice: first so that a/b and c/d mean anything at all, and again at the end to conclude $bd \neq 0$.
- $a, c \in \mathbb{Z}$ are needed only for the numerator’s integrality.

The step “ $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$ ” deserves a name: it is the statement that $\mathbb{Z}$ has **no zero divisors**. It is true but not automatic — in $\mathbb{Z}_6$, for instance, $2 \neq 0$ and $3 \neq 0$ yet $2 \cdot 3 = 0$. Had we been working over such a ring, the proof would break at

precisely that line, and the “field of fractions” construction would fail. Flagging the step is therefore not pedantry; it is where an unstated property of $\mathbb{Z}$ gets used.

1.7 Remark — Distinct Letters for Distinct Witnesses

The proof writes $x = a/b$ and $y = c/d$ with **four different letters**. As in Problem 2.7.5, existential instantiation must introduce a fresh name every time. Writing $x = a/b$ and $y = a/b$ would silently assume $x = y$ and prove only that $2x$ is rational — a strictly weaker claim.

The same care matters for the two denominators: reusing b for both would assume the fractions share a denominator, which is a genuine restriction even though every pair of rationals *can* be put over a common denominator. Proving the general case requires starting general.

1.8 Remark — This Is One Axiom Among Several

Closure under addition is one of the properties that make $\mathbb{Q}$ a **field**, and the companion results are proved the same way:

claim	key computation
$x + y \in \mathbb{Q}$	$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$
$x \cdot y \in \mathbb{Q}$	$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$
$-x \in \mathbb{Q}$	$-\frac{a}{b} = \frac{-a}{b}$
$x^{-1} \in \mathbb{Q}$ for $x \neq 0$	$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$, needing $a \neq 0$

Table 1.1

Every one is “unpack, compute, repack,” and every one ends by checking that the new denominator is a nonzero integer. The last row is the only one carrying an extra hypothesis,

since $x \neq 0$ is exactly what guarantees $a \neq 0$.

1.9 Remark — The Analogous Claim for Irrationals Is False

It is tempting to expect a matching theorem for the irrationals, but there is none: **the sum of two irrationals need not be irrational**. Take $x = \sqrt{2}$ and $y = -\sqrt{2}$, both irrational by Section 1.6, with $x + y = 0 \in \mathbb{Q}$. Or $x = \sqrt{2}$ and $y = 1 - \sqrt{2}$, summing to 1.

The asymmetry has a clean explanation: $\mathbb{Q}$ is closed under addition *because* it is defined by a constructive condition (“is a ratio of integers”), which the computation preserves. The irrationals are defined by a **negation** (“is not such a ratio”), and negatively-defined sets are rarely closed under anything. The nearest true statement is a mixed one — rational plus irrational is always irrational — which is proved by contradiction using precisely the closure theorem established here.